CareFlow - Project Documentation

Project: CareFlow: Clinical Pathway Process Mining | Intern: Yash Rahangdale | Role: Data Associate L1 Intern (Data Analytics) | Organisation: Infotact Solutions | Ref. No. ITS/DA/1393

1. Objective

Read a hospital event log, reconstruct the real clinical pathway, find bottlenecks, rework and deviations, predict long stays, and publish results as charts, reports and a dashboard using the industry-standard workflow: collect, clean, validate, analyse, visualise, document.

2. Data model

One row = one event. Minimum columns: case_id (patient journey), activity (step), timestamp. Optional: resource, department, wait_minutes, service_minutes, cost_inr. Case-level: age, gender, triage_level, admission_type, pathway, los_hours, total_cost_inr.

3. Database design (star schema)

Table	Type	Key columns
dim_patient	dimension	patient_id, age, gender
dim_activity	dimension	activity, activity_group
dim_department	dimension	department
dim_resource	dimension	resource, department
dim_date	dimension	date_key, day_name, month_name, is_weekend
fact_case	fact	case_id, los_hours, total_cost_inr, is_long_stay
fact_event	fact	event_id, case_id, event_index, activity, event_timestamp, wait_minutes

MySQL note: TIMESTAMP is a reserved word, so the column is named event_timestamp. event_index must be rebuilt with ROW_NUMBER() OVER (PARTITION BY case_id ORDER BY event_timestamp) - that ordering is what makes process mining possible.

4. Cleaning rules applied (audit logged in outputs/reports/cleaning_audit.json)

Rule	Reason
Drop rows without case_id / activity / timestamp	cannot be mined
Remove exact duplicates (case + activity + timestamp)	double-scanned events - 81 removed
Blank resource -> UNKNOWN	keep the event, do not lose the path
Negative wait / cost -> 0	physically impossible
service_minutes <= 0 -> median	robust to outliers
Cases < 3 or > 200 events dropped	too short / corrupt journeys

5. Methods

Process discovery: Directly-Follows Graph and trace-variant analysis; optional Inductive / Heuristics / Alpha miner plus token-based replay via pm4py. Conformance: token-based replay fitness per case and per variant with deviation typing. Analytics: KPI cards, bottleneck ranking by average wait, department cost, resource workload, weekday x hour heat map, LOS distribution. Predictive: binary classification of long stay (> 24 h) with leakage-free features, 5-fold cross-validation, ROC-AUC / F1 comparison and confusion matrix.

6. Reproducibility

```
pip install -r requirements.txt
python run_all.py
python src/connect_db.py --engine sqlite
jupyter notebook notebooks/
```

7. Limitations

The reference log is synthetic, so conformance fitness is 1.0 and the classifier scores high. On real data expect fitness 0.7-0.95 and long-stay AUC 0.75-0.90. The resource dimension is coarse, cost is activity-level, and the long-stay class is highly imbalanced (1.13 %).

Infotact Solutions confidential - prepared by Yash Rahangdale.